Hands-on AI

Quick GPT Demos

- Write and Revise a Story from 6 word prompt
  - Understands the context of the story
  - Adapts to different forms/styles
- Understand images
- Explain humor
- Create images and movies

GPT-4 visual input example, Chicken Nugget Map:

User: Can you explain this meme?



GPT-4: This meme is a joke that combines two unrelated things: pictures of the earth from space and chicken nuggets.
The text of the meme suggests that the image below is a beautiful picture of the earth from space. However, the image is actually of chicken nuggets arranged to vaguely resemble a map of the world.
The humor in this meme comes from the unexpected juxtaposition of the text and the image. The text sets up an expectation of a majestic image of the earth, but the image is actually something mundane and silly.

LLM Uses

- Rough-draft prose, art, code, music
- Finding and fixing errors in grammar, code, etc.
- Summarizing, revising
- Analyzing large content intelligently
- Brainstorming
- But… still requires human review and refinement.

Prompting an LLM

- “Prompt” can be much more than a question
- LLM operates on training *and prompt*
- Like long term memory and short term memory
- You can’t change the training, but you control the prompt
- Examples of long prompts
  - Documents to analyze
  - Detailed instructions
  - Example outputs: “one-shot” or “few-shot” prompt

Improving Prompts

- Set “expertise” context.
  - “You are an expert on Civil War history”
  - Tends to associate with relevant training.
- Break task into numbered list of steps.
  - LLMs love steps.
- Request multiple examples and pick best

AI General Vocabulary

- Artificial Intelligence
- Machine Learning
- Supervised learning, and training process
- Unsupervised learning
- Neural Network / Deep Learning
- Generative AI

LLM Vocabulary

- Large Language Model (LLM)
- Diffusion Model (DALL-E)
- Generative Pretrained Transformer (GPT)
- Parameters/weights
- Tokens and token limit
- Hallucinations